# Assignment 2: Attention Mechanisms and Transformers

## 1 Part A: Self-Attention From Scratch

We implemented scaled dot-product self-attention for the sentence `"Darth Vader is the villain"` (5 tokens). Each token was assigned a random 4-dimensional embedding, and three fixed (untrained) weight matrices $W_Q, W_K, W_V \in R^{4\times 3}$ projected them into query, key, and value vectors. Attention was computed as $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d_k})\,V$ with $d_k = 3$.

| | Darth | Vader | is | the | villain |
|---|---|---|---|---|---|
| Darth | 0.135 | 0.050 | 0.291 | 0.169 | 0.356 |
| Vader | 0.051 | 0.309 | 0.249 | 0.285 | 0.106 |
| is | 0.324 | 0.055 | 0.206 | 0.119 | 0.296 |
| the | 0.192 | 0.167 | 0.229 | 0.201 | 0.210 |
| villain | 0.142 | 0.388 | 0.144 | 0.208 | 0.118 |

Table 1: Self-attention weights (rows = queries, columns = keys). Each row sums to 1.

Since the projection matrices are random and untrained, the attention weights do not reflect semantic relationships. However, the mechanism is correct: each token's context vector is a weighted combination of all value vectors, where weights are determined by query–key compatibility. In a trained model, semantically related pairs like "villain" and "Vader" would receive high attention weights. Interestingly, "villain" already attends most strongly to "Vader" (0.388) purely by chance in this random initialisation.

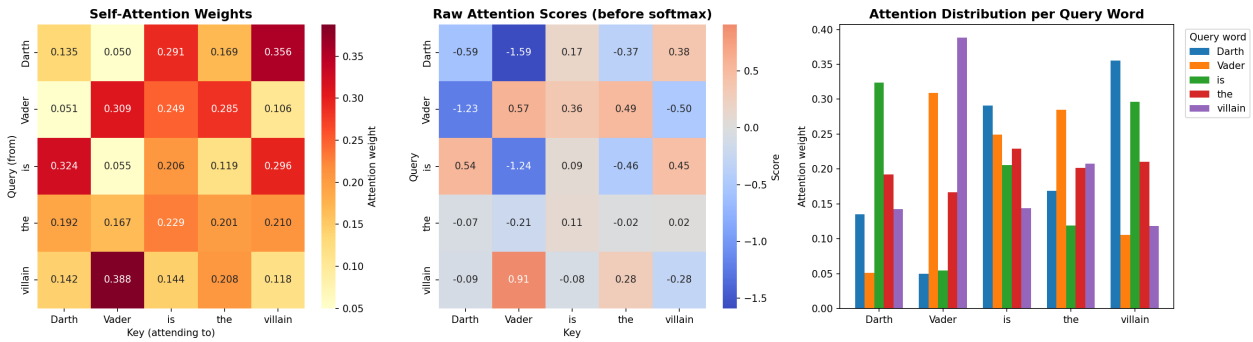


Figure 1: Left: attention weight heatmap. Centre: raw scores before softmax. Right: per-query attention distribution.

## 2 Part B: Transformer-Based Movie Chatbot

We used T5-small (60.5M parameters), a pretrained encoder-decoder Transformer, to build a simple movie QA chatbot. A small corpus of 6 movies (Star Wars, The Godfather, The Dark Knight, Inception, Titanic, Pulp Fiction) with IMDb-style plot summaries was prepared, yielding 24 QA pairs. Each input follows the format `question: ... context: ...`, and the model generates the answer as free text. Fine-tuning was intentionally minimal: 3 epochs with AdamW ($lr = 3 \times 10^{-4}$, batch size 4). The loss decreased from 0.592 to 0.303 but remained high, confirming the model did not converge.

| Question | Model Answer | Correct? |
|---|---|---|
| Who is the villain in Star Wars? | Christian Bale | × |
| What movie features Darth Vader? | The Galactic Empire | × |
| Who directed The Godfather? | Francis Ford Coppola | |
| Who plays the Joker in The Dark Knight? | Christian Bale | × |
| What happens to the Titanic? | It hits an iceberg and sinks | |
| Who stars in Inception? | Christian Bale | × |
| Who directed Pulp Fiction? | Quentin Tarantino | |
| Who is the villain in Titanic? | Christian Bale | × |
| What year was Inception released? | 2010 | |
| Who plays Mia Wallace in Pulp Fiction? | Uma Thurman | |

Table 2: Chatbot test results (5/10 correct).

The model achieved 50% accuracy. It answers factual questions well (directors, years, plot events) but shows a strong "default answer" bias toward "Christian Bale," suggesting it latched onto a frequent name during fine-tuning rather than learning the QA mapping properly.

## 3 Analysis

**Failures and hallucination.** The model hallucinates confidently: it answers "Christian Bale" for Star Wars, Inception, and Titanic—movies where Bale does not appear. This is a direct consequence of minimal fine-tuning on only 24 examples over 3 epochs. The model also struggles with answer-type disambiguation (returning "The Galactic Empire" instead of a movie title) and with questions that have no answer in the corpus ("Who is the villain in Titanic?").

**How attention helps in long contexts.** Self-attention allows every token to attend directly to every other token regardless of distance. This means the model can link "villain" in the question to "Darth Vader" hundreds of tokens away in the context without passing through intermediate hidden states (as RNNs must). Multi-head attention further enables different heads to capture different relationship types (syntactic, semantic, coreference) in parallel. This $O(1)$ path length between any two positions is the fundamental advantage over recurrence ($O(n)$ steps) and convolution ($O(\log n)$ layers).

**Dataset vs. model limitations.** On the dataset side: the corpus is tiny (6 movies, 24 QA pairs), each question phrasing is seen only once, and some queries have no answer in the text. On the model side: T5-small has limited capacity, 512-token input truncation cuts off part of the concatenated context, and 3 epochs of fine-tuning is insufficient to learn the task. With more data, more diverse question phrasings, and more training epochs, accuracy would improve substantially—but even with this minimal setup, the model demonstrates that pretrained Transformers can extract and generate answers from unstructured text.